



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

May 2008

Half Yearly Examination

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question

Total Marks – 60

- All Questions may be attempted
- Each Section should be handed up in a separate examination Booklet.
- Full Marks may not be awarded for careless or poorly set out work.

Examiner – *A.M. Gainford*

Section A
(Start a new booklet)

Question 1. (10 Marks)

- | | Marks |
|--|--------------|
| (a) Find the exact value (as a fraction in lowest terms) of:
$\frac{\frac{3}{4} - \frac{7}{24}}{1 + \frac{3}{4} \times \frac{7}{24}}$ | 2 |
| (b) Evaluate $\frac{\sqrt{3}+1}{\sqrt{3}-1}$, correct to two decimal places. | 2 |
| (c) Find the point of intersection of the lines $3x + 2y = 10$ and $4x + 3y = 13$. | 3 |
| (d) Simplify $x(2 - y) - y(2 - x)$. | 1 |
| (e) Find x and y if $x + y + \sqrt{x - y} = 10 + 2\sqrt{2}$. | 2 |

Question 2. (10 Marks)

- | | Marks |
|--|--------------|
| (a) Express $3.57 \times 10^7 \times 7.64 \times 10^6$ in scientific notation, correct to three significant figures. | 1 |
| (b) Find all the solutions of $ 5x - 2 = 3x + 4 $. | 2 |
| (c) Factorise: (i) $4x^2 + 4x - 3$
(ii) $x^2 - 3x + 2ax - 6a$
(iii) $1 + 27a^3$ | 3 |
| (d) Solve for x : $3x(x - 3) = 0$ | 2 |
| (e) Solve $2 - 3x \geq 8$, and graph the solution on the number line. | 2 |

Section B
(Start a new booklet.)

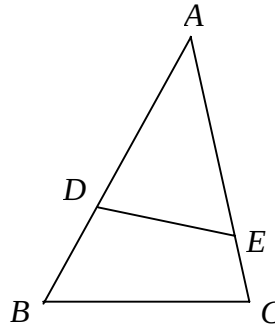
Question 3. (10 Marks)

Marks
3

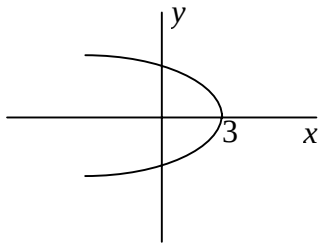
- (a) In the diagram at right $AD = 8\text{cm}$,
 $DB = 7\text{cm}$, $AE = 10\text{cm}$, $EC = 2\text{cm}$,
 $BC = 21\text{cm}$.

(i) Copy the diagram to your answer sheet.

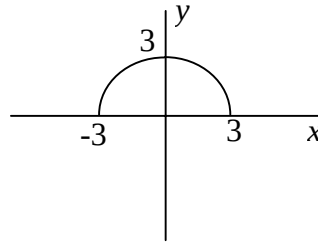
(ii) Prove that $\triangle ADE$ is similar to $\triangle ACB$, and hence find the length of DE (giving full reasons).



- (b) Figure A



- Figure B



2

Indicate which of these graphs represents a function, and state its domain.

- (c) For the points $A(-1, 2)$ and $B(2, 4)$:

5

- (i) Find the coordinates of M , the midpoint of AB .
- (ii) Write the equation of the line AB .
- (iii) Find the equation of the line through M perpendicular to AB .

Section continued overleaf.

Question 4. (10 Marks)

Marks

1

(a) Express $\frac{5\pi}{9}$ radians in degrees.

1

(b) Express $0.\dot{1}3\dot{7}$ as a common fraction in lowest terms.

2

(c) State the exact value of:

(i) $\cos 135^\circ$

(ii) $\operatorname{cosec} 330^\circ$

6

(d) Solve for x:

(i) $x^4 - 25x^2 + 144 = 0$

(ii) $\frac{3}{x-1} > 3$

(iii) $\frac{1}{|2-x|} < 2$

Section C
(Start a new booklet)

Question 5. (10 Marks)

- | | Marks |
|--|--------------|
| (a) Show that $\tan \theta + \cot \theta = \operatorname{cosec} \theta \sec \theta$. | 1 |
| (b) Find the perpendicular distance of the point $(1, -2)$ from the line $x - 2y = -3$. | 1 |
| (c) Find the equation of the line through the intersection of the lines $x - 2y + 1 = 0$ and $3x - y - 2 = 0$, and passing through $(3, 1)$. | 2 |
| (d) Find the equation of the locus of all points equidistant from the fixed points $A(2, -1)$ and $B(-3, 2)$. | 2 |
| (e) On separate diagrams, sketch the graphs of the following: | 4 |
| (i) $y = 1 - 2^x$ | |
| (ii) $y = \frac{ x }{x}$ | |

Section continued overleaf.

Question 6. (10 Marks)**Marks**

(a) Show that the function $f(x) = \frac{2^x + 2^{-x}}{1 - x^2}$ is an even function.

2

(b) Solve the simultaneous equations:

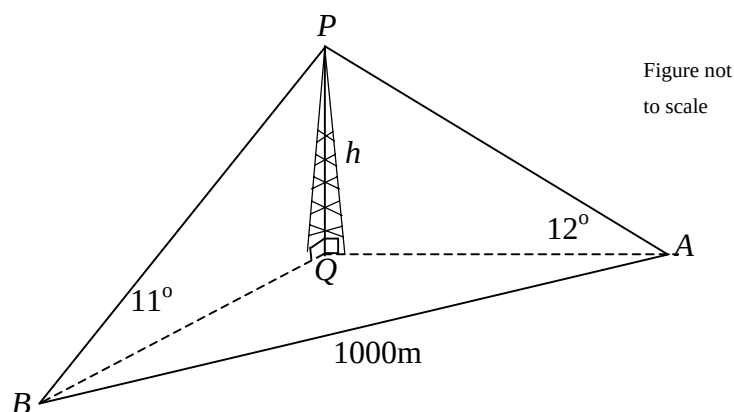
3

$$5a - 2b + 6c = 3$$

$$6a + 4b - 4c = 0$$

$$3a - 4b + 8c = 3$$

(d)

5

The angle of elevation of a tower PQ of height h metres at a point A due east of it is 12° . From another point B , the bearing of the tower is 051°T and the angle of elevation is 11° . The points A and B are 1000 metres apart and on the same level as the base Q of the tower.

(i) Show that $\angle AQB = 141^\circ$.

(ii) Consider the triangle APQ and show that $AQ = h \tan 78^\circ$.

(iii) Find a similar expression for BQ .

(iv) Use the cosine rule in the triangle AQB to calculate h to the nearest metre.

This is the end of the paper.

QUESTION 1

$$(a) \frac{11}{24} \div \frac{39}{32}$$

$$= \frac{44}{117}$$

$$(b) 3.73$$

$$(c) \textcircled{1} 3x + 2y = 10$$

$$\textcircled{2} 4x + 3y = 13$$

$$\textcircled{1} 9x + 6y = 30$$

$$\textcircled{2} 8x + 6y = 26$$

$$x = 4$$

$$\textcircled{1} 12 + 2y = 10$$

$$y = -1$$

$$(4, -1)$$

$$(d) \begin{array}{l} 2x - xy - 2y + xy \\ 2x - 2y \end{array}$$

$$(e) \begin{array}{l} x + y = 10 \\ \sqrt{x-y} = 2\sqrt{2} = \sqrt{8} \\ x - y = 8 \\ 2x = 18 \\ x = 9 \\ y = 1 \end{array}$$

QUESTION 2

$$(a) 2.73 \times 10^{14}$$

$$(b) 5x - 2 = 3x + 4$$

$$2x = 6$$

$$x = 3$$

or

$$5x - 2 = -3x - 4$$

$$8x = -2$$

$$x = -\frac{1}{4}$$

$$x = 3, -\frac{1}{4}$$

$$(c) \textcircled{1} (2x-1)(2x+3)$$

$$\textcircled{ii} x(x-3) + 2a(x-3)$$

$$(x+2a)(x-3)$$

$$\textcircled{iii} (1+3a)(1-6a+9a^2)$$

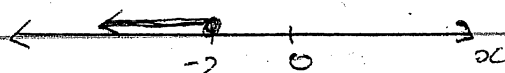
$$(d) 3x(x-3) = 0$$

$$x = 0, x = 3$$

$$(e) 2 - 3x \geq 8$$

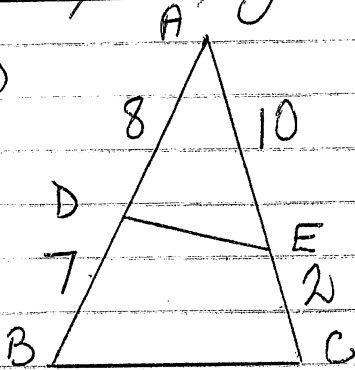
$$-3x \geq 6$$

$$x \leq -2$$



VR11 2008 Half Yearly Maths Extension (Continuers)

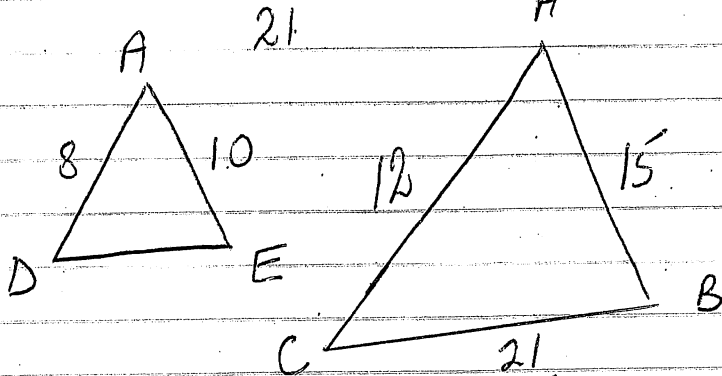
(3) (a) (i)



~~12~~

10

(ii)



$$\frac{AC}{AD} = \frac{12}{8} = 1.5$$

$$\frac{AB}{AE} = \frac{15}{10} = 1.5$$

} sides in same ratio

$(1\frac{1}{2})$

$\hat{DAE} = \hat{CAB}$ equal angle, same angle etc

So $\triangle ADE \parallel \triangle ACB$ sides in common ratio and included angle test.

$$\therefore \frac{BC}{DE} = 1.5 \quad (\text{sides in same ratio})$$

$$\text{so } \frac{21}{DE} = 1.5 \quad \text{So } DE = 14 \text{ cm}$$

(1)

(b) Figure B. Domain $-3 \leq x \leq 3$ (2)

(c) (i) $M\left(\frac{-1+2}{2}, \frac{2+4}{2}\right) = \left(\frac{1}{2}, 3\right)$ (1)

(ii) $m = \frac{4-2}{2+1} = \frac{2}{3}$ using $(y-y_1) = m(x-x_1)$
 $y-2 = \frac{2}{3}(x+1)$
 $y-2 = \frac{2}{3}x + \frac{2}{3}$

$$\text{So } y = \frac{2}{3}x + 2\frac{2}{3}$$

(2)

$$\text{Or } 3y - 6 = 2x + 2$$

$$0 = 2x - 3y + 8$$

$$(iii) m = -\frac{3}{2} \quad m(\frac{1}{2}, 3)$$

$$(y-3) = -\frac{3}{2}(x-\frac{1}{2})$$

$$y-3 = -\frac{3x}{2} + \frac{3}{4}$$

$$y = -\frac{3x}{2} + 3\frac{3}{4} \quad \text{or } 2y-6 = -3x + \frac{3}{2}$$

$$4y-12 = -6x+3$$

$$6x+4y-15=0$$

(2)

$$(4) (a) \frac{5 \times 180}{9} = 100^\circ \quad (1)$$

10

$$(b) \text{ let } x = 0.13737 \dots$$

$$100x = 13.73737$$

$$99x = 13.6$$

$$x = \frac{13.6}{99} = \frac{136}{990} = \frac{68}{495} \quad (1)$$

$$(c) (i) \cos 135^\circ = -\cos 45^\circ = -\frac{1}{\sqrt{2}} \quad (1)$$

$$(ii) \operatorname{cosec} 330^\circ = -\operatorname{cosec} 30^\circ = -\frac{1}{\sin 30^\circ} = -\frac{1}{\frac{1}{2}} = -2$$

$$(d) (i) x^4 - 25x^2 + 144 = 0$$

$$\text{let } w = x^2$$

$$w^2 - 25w + 144 = 0$$

$$(w-9)(w-16) = 0$$

$$w=9 \text{ and } w=16$$

$$\text{So } x^2 = 9, x^2 = 16$$

$$x = \pm 3, \pm 4$$

(2)

④ (d) (ii) $\frac{3}{(x-1)} > 3 \quad x \neq 1$

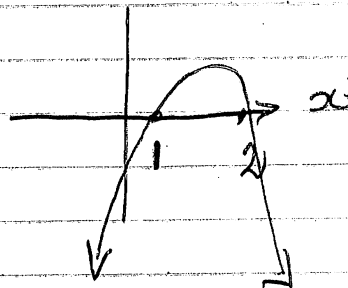
$$\frac{(x-1)^2 \times 3}{(x-1)} > 3(x-1)^2$$

$$3(x-1) - 3(x-1)^2 > 0$$

$$3(x-1)[1-(x-1)] > 0$$

$$3(x-1)[2-x] > 0$$

soln $1 < x < 2$ ②



(iii) $\frac{1}{|2-x|} < 2$

$x \neq 2$. take reciprocals.

$$|2-x| > \frac{1}{2}$$

$$2-x > \frac{1}{2}$$

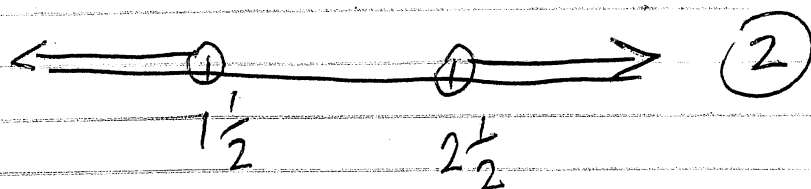
$$2-x < -\frac{1}{2}$$

$$-x > -1\frac{1}{2}$$

$$-x < -2\frac{1}{2}$$

$$x < 1\frac{1}{2}$$

$$x > 2\frac{1}{2}$$



SECTION C

Question 5

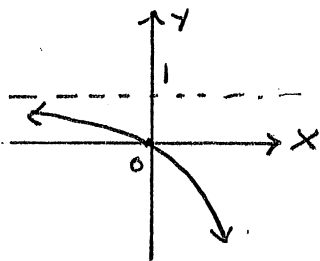
$$\begin{aligned}
 (a) \text{ LHS} &= \tan \theta + \cot \theta \\
 &= \frac{s}{c} + \frac{c}{s} \\
 &= \frac{c^2 + s^2}{cs} \\
 &= \frac{1}{sc} \\
 &= \frac{1}{s} \cdot \frac{1}{c} \\
 &= \operatorname{cosec} \theta \cdot \sec \theta \\
 &= \text{RHS}
 \end{aligned}$$

$$(b) \frac{|1(1) - 2(-2) + 3|}{\sqrt{(1)^2 + (-2)^2}} = \frac{8}{\sqrt{5}}$$

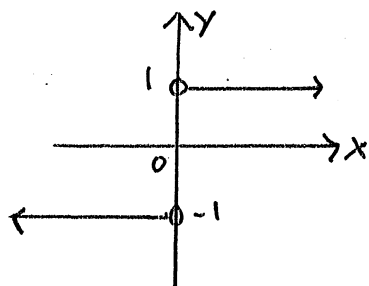
$$(d) \text{ PA} = \text{PB}$$

$$\begin{aligned}
 \sqrt{(x-2)^2 + (y+1)^2} &= \sqrt{(x+3)^2 + (y-2)^2} \\
 5x - 3y + 4 &= 0
 \end{aligned}$$

(e) (i)



(ii)



QUESTION 6

$$\begin{aligned}
 (a) \quad f(x) &= \frac{2^x + 2^{-x}}{1 - x^2} \\
 f(-x) &= \frac{2^{-x} + 2^{-(-x)}}{1 - (-x)^2} \\
 &= \frac{2^{-x} + 2^x}{1 - x^2} = f(x) \\
 \Rightarrow f(x) &\text{ even}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad 10a - 4b + 12c &= 6 \quad \text{--- (A)} \\
 6a + 4b - 4c &= 0 \quad \text{--- (B)} \\
 3a - 4b + 8c &= 3 \quad \text{--- (C)}
 \end{aligned}$$

$$(A) + (B) \Rightarrow 16a + 8c = 6 \quad \text{--- (D)}$$

$$(B) + (C) \Rightarrow 9a + 4c = 3 \quad \text{--- (E)}$$

$$\therefore a = 0, b = c = \frac{3}{4}$$

$$\begin{aligned}
 (c) \quad \text{Required line} \\
 (x - 2y + 1) + k(3x - y - 2) &= 0 \\
 (3 - 2 + 1) + k(9 - 1 - 2) &= 0
 \end{aligned}$$

$$\Rightarrow k = -\frac{1}{3}$$

$$\therefore (x - 2y + 1) - \frac{1}{3}(3x - y - 2) = 0$$

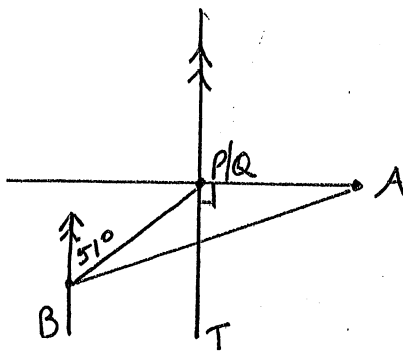
$$3(x - 2y + 1) - (3x - y - 2) = 0$$

$$\therefore -5y + 5 = 0$$

$$y = 1$$

(c)

(i)

2D DIAGRAM
(TOP VIEW)

$$\hat{AQB} = 90^\circ + \hat{BQT}$$

$$= 90^\circ + 51^\circ$$

$$\hat{AQB} = 141^\circ$$

$$(ii) \tan \hat{QPA} = \frac{AQ}{h}$$

$$\tan 78^\circ = \frac{AQ}{h}$$

$$\Rightarrow AQ = h \tan 78^\circ$$

(iii) In $\triangle BPQ$

$$\tan 11^\circ = \frac{h}{BQ}$$

$$BQ = \frac{h}{\tan 11^\circ} = h \cot 11^\circ$$

$$\text{or } BQ = h \tan 79^\circ$$

$$(iv) \quad 1000^2 = h^2 \tan^2 79^\circ + h^2 \tan^2 78^\circ - 2h \tan 79^\circ h \tan 78^\circ$$

$$= h^2 [\tan^2 79^\circ + \tan^2 78^\circ - 2 \tan 79^\circ \tan 78^\circ]$$

$$\Rightarrow h = \frac{1000}{[\quad]^{\frac{1}{2}}}$$

$$h = 108 \text{ metres}$$